N-O Cool-chic: reconcile fast encoding with lightweight decoding for neural image compression

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Context & contributions

Cool-chic is an overfitted neural image codec. A lightweight decoder is learnt and sent for each image alongside a latent representation.

With 2 000 multiplications per decoded pixel, it is competitive with VVC. However, the encoding time associated with the overfitting procedure may be prohibitively long for real-time applications.

Contributions: fast encoding for Cool-chic

- Bypass the overfitting procedure
- Complement the decoder with an encoder network

Cool-chic & Neural Representation

An image $\mathbf{x}$ is represented by multi-resolution latent grids $\hat{\mathbf{y}}$. The upsampling $f_{\boldsymbol{\upsilon}}$ and the synthesis $f_{\boldsymbol{\theta}}$ NNs reconstruct the image.

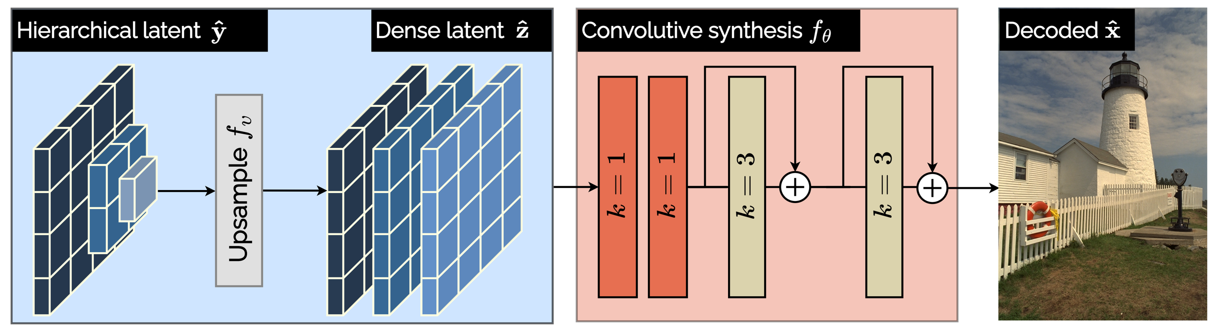


Figure 1. Synthesizing an image $\hat{\mathbf{x}}$ from a latent representation $\hat{\mathbf{y}}$

The synthesis, the upsampling and the latent grids are overfitted *i.e.* learnt for each image to minimize the distortion of the reconstruction.

$$\hat{\mathbf{y}}, \boldsymbol{\theta}, \boldsymbol{\upsilon} = \arg\min ||\mathbf{x} - f_{\boldsymbol{\theta}}(f_{\boldsymbol{\upsilon}}(\hat{\mathbf{y}}))||^2 = \arg\min \mathrm{D}(\mathbf{x}, \hat{\mathbf{x}}) \quad (1)$$

Latent grids entropy coding

The latent grids $\hat{\mathbf{y}}$ are sent using entropy coding to reduce their rate.

The NN $f_{\boldsymbol{\psi}}$ models the conditional distribution of the latent grids as a parameterized Laplace $p_{\boldsymbol{\psi}}$. It drives the entropy coding algorithm, exploiting the spatial redundancies of the latents to reduce their rate.

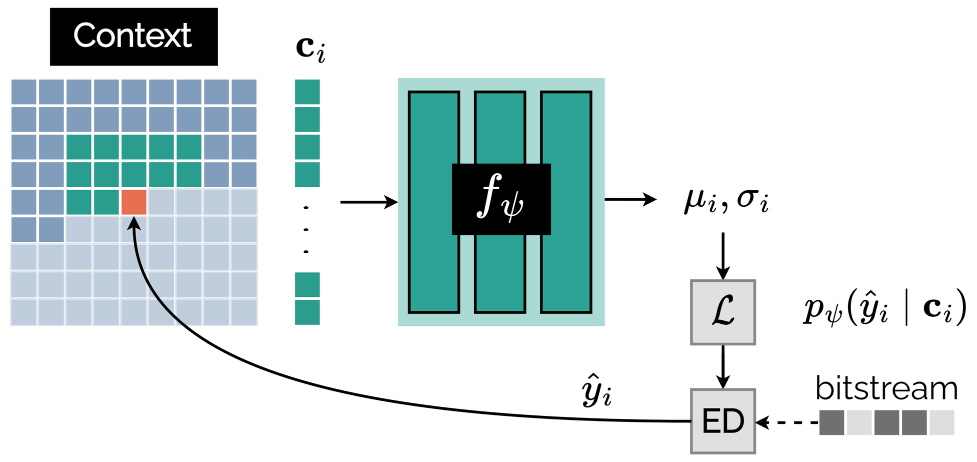
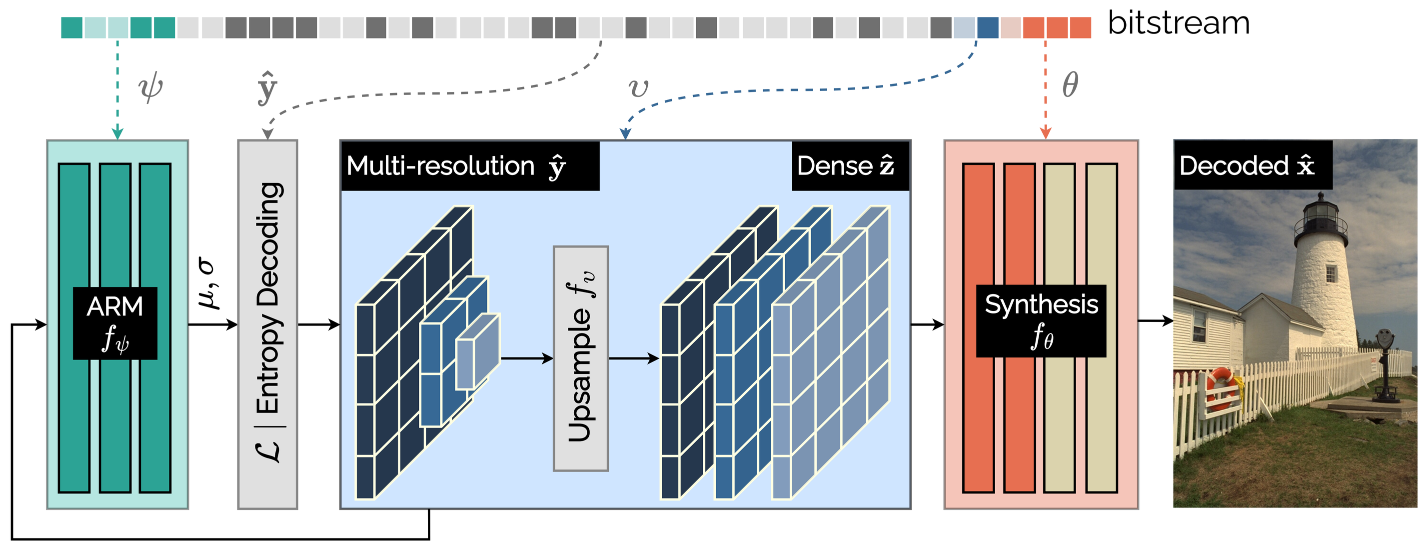


Figure 2. Entropy decoding of the latent grids $\hat{\mathbf{y}}$ using the probability model $p_{\boldsymbol{\psi}}$. $\mathcal{L}$ stands for Laplace distribution.

The auto-regressive NN and the latent representation are learnt for each image to minimize the latent grids rate.

$$\hat{\mathbf{y}}, \boldsymbol{\psi} = \arg\min \sum_i -\log_2 p_{\boldsymbol{\psi}}(\hat{y}_i \mid \mathbf{c}_i) = \arg\min \mathrm{R}(\hat{\mathbf{y}}) \quad (2)$$

Decoding an image



3-step decoding process

1. Decode NNs weights $\boldsymbol{\theta}, \boldsymbol{\psi}, \boldsymbol{\upsilon}$ from the bitstream
2. Decode the latent grids $\hat{\mathbf{y}}$ using the probability model $p_{\boldsymbol{\psi}}$
3. Upsample the latent grids and synthesize the decoded image $\hat{\mathbf{x}} = f_{\boldsymbol{\theta}}(f_{\boldsymbol{\upsilon}}(\hat{\mathbf{y}}))$

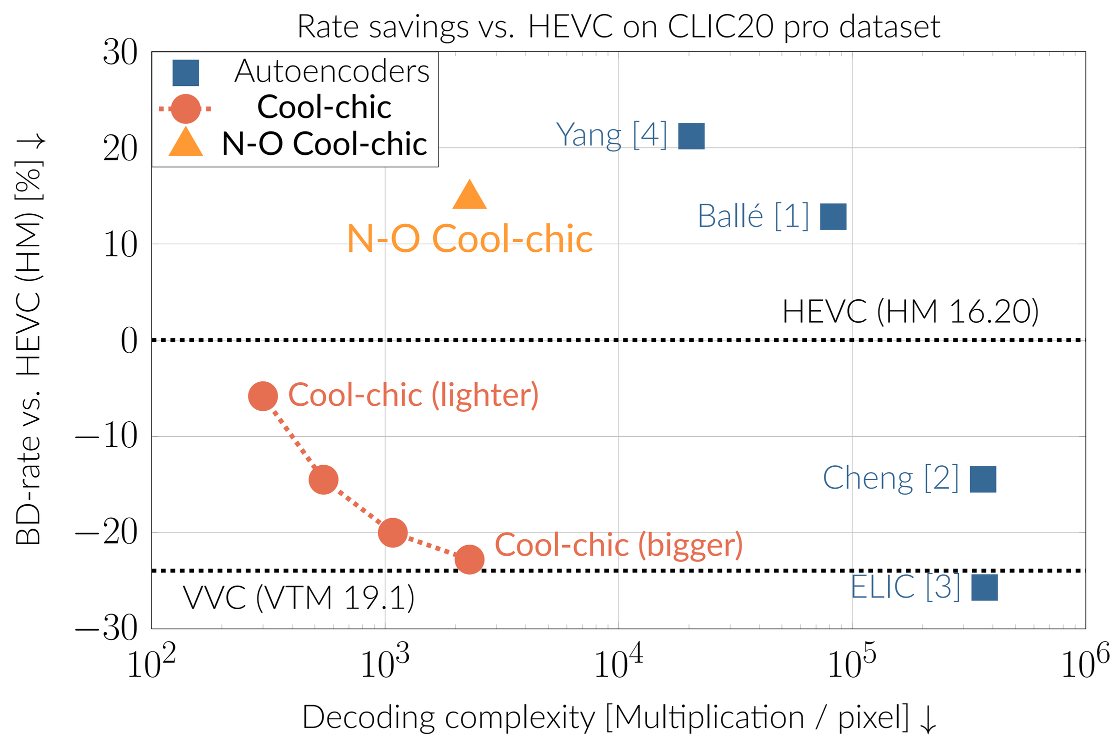
Figure 3. Cool-chic decoding process. $\mathcal{L}$ stands for Laplace distribution.

Encoding an image

Combining the equations (1) and (2) gives the rate-distortion objective optimized by Cool-chic when encoding an image.

$$\hat{\mathbf{y}}, \boldsymbol{\psi}, \boldsymbol{\theta}, \boldsymbol{\upsilon} = \arg\min \mathrm{D}(\mathbf{x}, \hat{\mathbf{x}}) + \lambda \mathrm{R}(\hat{\mathbf{y}}) \;\; \textit{i.e.} \;\; \text{Encoding an image} \Leftrightarrow \text{Finding the best parameters } \hat{\mathbf{y}}, \boldsymbol{\psi}, \boldsymbol{\theta}, \boldsymbol{\upsilon} \quad (3)$$

Rate-distortion performance



Cool-chic offers compelling rate-distortion performance.

Ballé [1]	HEVC (HM)	Cheng [2]	VVC (VTM)	ELIC [3]
- 30.4 %	- 22.7 %	-10.1 %	+ 0.8 %	+ 3.6 %

Table 1. Rate savings of Cool-chic against several codecs (CLIC20 pro dataset)

Cool-chic decoder is **lightweight** and **performant**

- Better than HEVC with 300 multiplications / decoded pixel
- Competitive with VVC for 2300 multiplications / decoded pixel
- Better than Cheng *et al.* with a decoder 1000x less complex

Encoding complexity

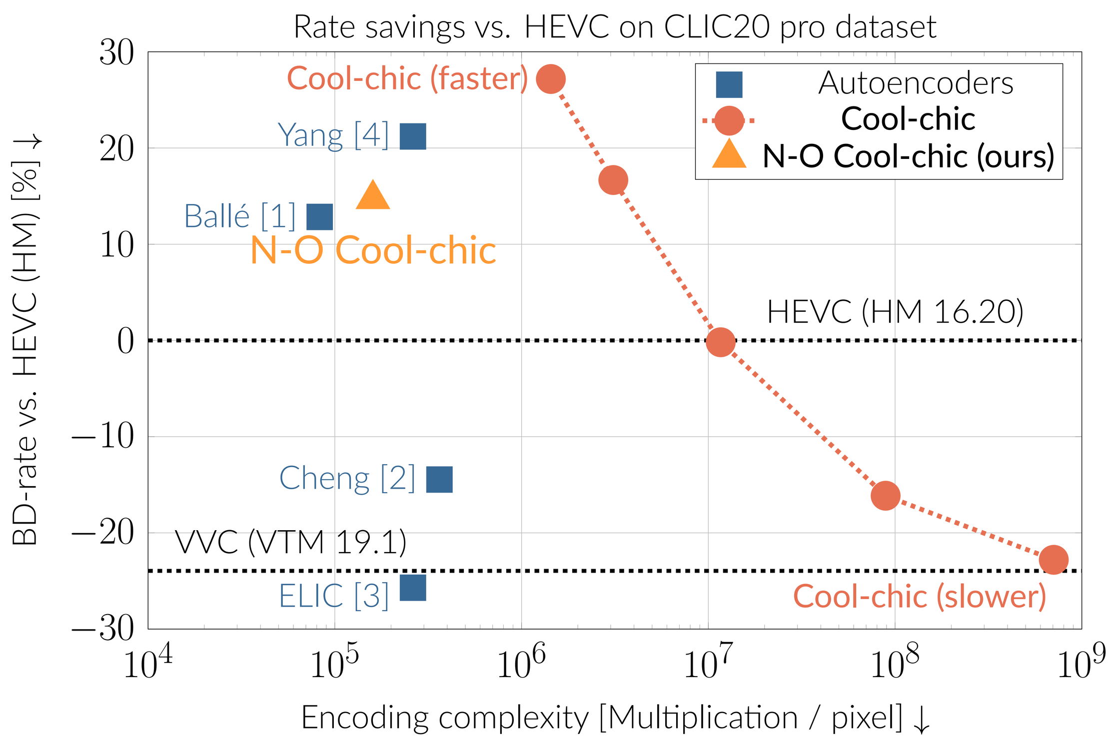
Cool-chic has a high encoding complexity.

Cool-chic	Cheng [2]	ELIC [3]	N-O (ours)	Ballé [1]
708000	350	262	160	80

Table 2. Encoding complexity (kMAC / pixel)

Cool-chic **iterative encoding** through gradient descent is computationally **expensive**.

- Encoding is 100x to 1000x more complex than autoencoders
- Can be reduced by decreasing the training time
- Beyond 10x reduction, performance drops significantly



Non-overfitted Cool-chic

To reduce the encoding time, the overfitting process is bypassed by training a non-overfitted (N-O) Cool-chic, sharing identical parameters for all images.

Encoding an image relies on an additional analysis transform $f_{\boldsymbol{\alpha}}$ generating the latent $\hat{\mathbf{y}}$ from the input image $\mathbf{x}$.

i.e. Encoding an image $\Leftrightarrow$ A single forward pass in the analysis $f_{\boldsymbol{\alpha}}$

Analysis transform

To produce a set of $L = 7$ multi-resolution latent grids,

- A series of L residual blocks progressively downsample the input image $\mathbf{x}$ and extract relevant features for the different resolutions
- After each downsampling step, a 1×1 convolution merges the $C = 64$ features into the l-th latent $\hat{\mathbf{y}}_l$

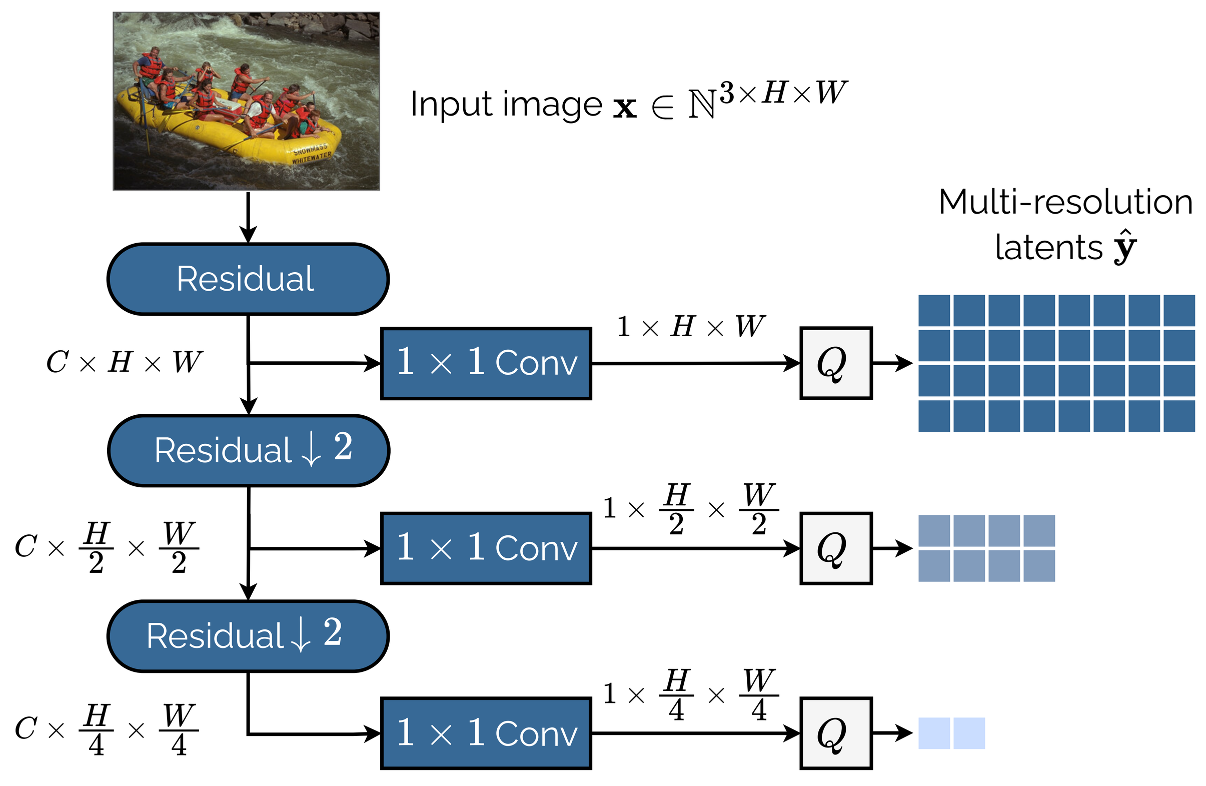


Figure 4. Encoding an image $\mathbf{x}$ with the analysis $f_{\boldsymbol{\alpha}}$. Q stands for quantization.

Results

N-O Cool-chic offers the benefits of both low encoding and decoding complexity. The encoding complexity is reduced 1000x compared to overfitted Cool-chic while maintaining competitive performance.

- On par with Ballé [1], with similar encoding complexity but with a decoder that is 30x less complex
- Better than Yang [4], while having a decoder that is 8x less complex

Conclusion & future work

- Cool-chic is an overfitted image codec competitive with VVC.
- The encoding time associated with the overfitting procedure is impracticable for real-time applications.
- Can be reduced 1000x by replacing the overfitting with a simple forward pass, while preserving most of the performance.
- Overfitting is still required to achieve best results.

References

[1] Ballé et al. Variational image compression with a scale hyperprior. In *6th International Conference on Learning Representations, ICLR 2018*.

[2] Cheng et al. Learned image compression with discretized gaussian mixture likelihoods and attention modules. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR*.

[3] Dailan He, Ziming Yang, Weikun Peng, Rui Ma, Hongwei Qin, and Yan Wang. Elic: Efficient learned image compression with unevenly grouped space-channel contextual adaptive coding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.

[4] Yibo Yang and Stephan Mandt. Computationally-efficient neural image compression with shallow decoders. In *IEEE/CVF International Conference on Computer Vision, ICCV 2023*.